

Ekman Spirals with Variable Eddy Viscosity

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Author: Fredrik Bergelv (fredrik.bergelv@live.se)

Supervisor: Jonas Nycander

1 Background

1.1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and current in the deep ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is often at an angle from the flow aloft (or down under in the ocean) by an angle of more or less 45° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft (or current below). This is known as the Ekman transport and is crucial for studying both meteorology and oceanography.

However, this derivation with a surface angle of 45° assumes a constant viscosity in the medium. However, this is a simplification of reality, where a turbulent boundary layer is present. Thus, in reality the viscosity should vary with altitude. This would then vary the surface angle from 45° , and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done both theoretically and numerically and will also be compared to model data. This will give a deeper insight into the Ekman spiral and provide insight into when a surface angle different from 45° can be observed.

1.2 Theory

1.2.1 problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right), \\ fu = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right), \end{cases} \quad (1a)$$

$$(1b)$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will thus approach geostrophic balance as,

$$-fv_g = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2)$$

$$fu_g = -\frac{1}{\rho} \frac{\partial p}{\partial y}. \quad (3)$$

Subtracting the geostrophic balance from Equation 1 gives

$$\begin{cases} -f(v - v_g) = \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right), \\ f(u - u_g) = \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right). \end{cases} \quad (4a)$$

$$(4b)$$

This problem can be formulated with one equation if one introduces a complex velocity defined as

$$U \equiv u + iv, \quad (5)$$

and a complex geostrophic velocity as $U_g = u_g + iv_g$. Using Equation 4 with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\frac{\partial}{\partial z} \left(\nu \frac{\partial U}{\partial z} \right) = if(U - U_g). \quad (6)$$

1.2.2 The classical solution

The classical solution to the Ekman equation is when we have a constant viscosity with altitude as $\nu = \text{constant}$.

2 Methodology

3 Results

4 Discussion

5 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

References

V. Walfrid Ekman. On the Influence of the Earth's Rotation on Ocean-Currents. *Arkiv för matematik, astronomi och fysik*, 2(11):1–52, May 1905.